

KABA HE, China

WMO: 510530

Lat: 48.0516N

Lon: 86.4057E

Elev: 534

StdP: 95.07

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.3	-27.0	-33.6	0.2	-30.2	-30.1	0.2	-26.9	11.9	-10.2	10.9	-9.2	1.4	290	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.7	19.0	31.2	18.4	29.6	17.9	20.5	29.7	19.6	28.5	18.9	27.4	2.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	13.3	24.1	16.4	12.5	23.3	15.6	11.8	23.0	61.6	29.9	58.6	28.4	56.0	27.6	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.7	7.7	-31.0	35.9	3.8	1.9	-33.8	37.2	-36.0	38.3	-38.2	39.4	-41.0	40.7		
			-31.2	22.2	3.8	1.8	-33.9	23.5	-36.1	24.5	-38.3	25.5	-41.0	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-14.7	-11.2	-2.4	9.0	15.6	21.3	22.9	21.2	15.2	7.7	-2.3	-11.1
	DBStd	14.25	6.59	6.37	7.14	5.61	4.47	3.53	2.94	3.37	4.39	4.99	6.98	5.94
	HDD10.0	2968	765	592	385	81	8	0	0	0	8	104	369	656
	HDD18.3	4880	1023	826	642	279	107	14	2	12	112	331	619	914
	CDD10.0	1518	0	0	2	52	181	338	401	347	164	32	1	0
	CDD18.3	388	0	0	0	1	23	102	145	101	17	0	0	0
	CDH23.3	3661	0	0	0	18	239	938	1323	977	164	2	0	0
	CDH26.7	1257	0	0	0	2	56	323	498	345	34	0	0	0
Wind	WSAvg	3.2	4.1	3.7	3.4	3.6	3.3	2.5	2.1	2.2	2.5	2.9	3.6	4.1
Precipitation	PrecAvg	176	6	6	7	16	22	19	22	16	15	19	18	9
	PrecMax	295	20	23	23	35	68	69	60	60	47	40	48	26
	PrecMin	90	0	1	0	0	0	2	0	0	1	2	4	1
	PrecStd	55	5	5	6	11	17	18	15	16	9	10	11	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.2	2.0	16.6	25.5	30.8	33.7	35.5	34.3	30.2	22.7	13.5	1.8
		MCWB	-2.2	-1.7	7.5	12.4	16.0	19.3	20.7	19.2	16.3	10.3	5.9	-1.9
	2%	DB	-3.3	-0.3	13.1	22.7	28.3	31.8	32.9	32.1	27.2	19.6	10.8	-0.3
		MCWB	-5.1	-2.7	6.2	11.1	14.7	18.6	19.5	18.5	14.8	9.0	4.1	-2.7
	5%	DB	-5.0	-2.0	10.1	20.2	26.2	30.1	31.2	30.2	25.1	17.3	8.5	-2.3
		MCWB	-6.8	-4.3	4.6	9.9	13.9	17.9	19.0	18.0	13.7	8.5	3.2	-4.1
	10%	DB	-6.7	-3.7	7.1	17.9	24.1	28.5	29.5	28.6	23.1	15.1	6.2	-3.9
		MCWB	-8.2	-5.6	3.1	8.8	12.9	17.3	18.5	17.4	12.8	7.7	2.1	-5.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.9	-1.1	8.5	13.5	17.4	20.9	22.6	21.2	17.3	11.8	7.5	-0.7
		MCDB	-0.9	1.3	14.0	22.3	27.5	31.1	32.1	30.8	27.2	17.9	11.7	1.1
	2%	WB	-5.1	-2.6	6.6	11.9	15.8	19.8	20.9	19.8	16.0	10.3	5.1	-2.5
		MCDB	-3.5	-0.4	12.4	20.0	25.2	29.3	29.9	29.2	25.3	16.8	9.1	-0.6
	5%	WB	-6.7	-4.1	5.0	10.8	14.9	18.9	20.0	18.9	14.7	9.3	3.6	-4.0
		MCDB	-5.2	-2.1	9.7	18.3	23.7	27.8	28.6	28.2	23.0	15.7	7.7	-2.3
	10%	WB	-8.1	-5.5	3.2	9.7	13.9	18.0	19.3	18.1	13.6	8.3	2.3	-5.5
		MCDB	-6.7	-3.9	7.0	16.3	22.4	26.7	27.5	26.7	21.2	14.3	6.1	-4.0
Mean Daily Temperature Range	5% DB	MDBR	6.1	7.0	8.4	11.0	11.5	11.3	11.0	11.9	11.5	9.0	6.1	5.3
		MCDBR	5.9	7.7	11.4	13.8	13.8	13.2	13.1	13.8	13.9	12.2	8.6	5.2
		MCWBR	4.8	6.0	6.5	6.3	5.5	5.3	5.4	5.8	6.1	5.8	4.9	4.0
	5% WB	MCDBR	5.9	7.3	10.7	12.4	12.0	11.7	11.5	12.4	12.4	10.8	8.1	5.0
		MCWBR	5.0	5.8	6.4	6.0	5.1	5.0	4.9	5.6	6.0	5.5	5.0	4.0
Clear-Sky Solar Irradiance	taub	0.292	0.321	0.365	0.409	0.394	0.401	0.404	0.380	0.344	0.341	0.318	0.294	
	taud	2.114	2.073	2.073	2.070	2.176	2.236	2.252	2.289	2.340	2.248	2.163	2.093	
	Ebn at Noon	750	812	836	836	867	861	853	859	859	788	710	687	
	Edh at Noon	87	115	136	151	141	134	130	120	103	96	83	78	
All-Sky Solar Radiation	RadAvg	1.72	2.79	4.22	5.37	6.54	6.82	6.53	5.92	4.61	2.75	1.57	1.28	
	RadStd	0.11	0.12	0.23	0.31	0.36	0.37	0.39	0.26	0.34	0.27	0.16	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page